

Notes: Maksimovic, Phillips, and Prabhala (JFE, 2011)
Post-merger restructuring and the boundaries of the firm

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1 Big picture

- **Question.** What happens to the real assets of target firms after mergers? The paper asks whether acquirers passively absorb target assets or actively redraw firm boundaries by keeping, selling, and closing plants.
- **Setting.** The authors study U.S. manufacturing mergers announced from 1981 to 2000 and track target and acquirer plants using plant-level Census/LRD data.
- **Core empirical fact.** Post-merger restructuring is large and fast. Within three years, acquirers keep only about 54% of target plants, sell 27%, and close about 19%.
- **Main theoretical contrast.** The paper contrasts agency or empire-building views of mergers with a comparative-advantage view of firm boundaries. If empire building dominates, acquirers should keep too many assets and should not systematically improve the assets they retain. If comparative advantage dominates, acquirers should keep assets that fit their capabilities and sell or close the rest.
- **Core mechanism.** Acquirers are more likely to retain target plants when the plants are related to the acquirer's main divisions, when the plants are central to the target, and when the acquirer has high productivity in peripheral businesses.
- **Performance result.** Retained target plants improve in productivity and margins, while sold plants do not show comparable improvement.
- **Financing result.** Method of payment matters only weakly. Financial constraints matter more: high leverage, low cash, and non-dividend status are associated with more asset sales.
- **Economic takeaway.** Mergers are not the end of boundary formation. They are an initial step that triggers a short-run reallocation process. The evidence is more consistent with economically sensible comparative-advantage restructuring than with pure empire building.

2 Incremental contribution of the paper

- **From long-run divestitures to short-run plant-level restructuring.** Earlier work studies whether acquired firms are eventually divested over long horizons. This paper studies the immediate three-year restructuring after mergers at the plant level.
- **A direct test of firm-boundary theories.** The paper uses plant-level data to test whether firm boundaries shift in a way consistent with organizational capability and comparative advantage.
- **A data contribution.** The authors connect SDC merger data with the Census Bureau Longitudinal Research Database, allowing individual target plants to be tracked after they are absorbed into acquirers.
- **Disposition and performance together.** The paper does not only ask whether plants are kept, sold, or closed. It also asks whether the performance path differs for kept versus sold plants.
- **A marginal-skill contribution.** The paper emphasizes that acquirer average profitability is not enough. What matters for retention is the acquirer's skill at the margin, especially productivity in peripheral divisions.

- **A financing-side extension.** The paper adds method of payment and financial-constraint variables to a real-assets view of mergers.
- **An agency-boundary contribution.** The paper tests repeat acquirers and state anti-takeover laws. It finds little evidence that the real-asset side of merger outcomes is dominated by empire building.
- **A practical M&A contribution.** The paper shows that full-firm acquisitions often contain unwanted assets. Acquirers buy portfolios and then quickly reshape them.

3 Data and measurement

3.1 Merger sample

- The initial merger sample comes from the SDC mergers and acquisitions database.
- The sample includes U.S. targets in completed mergers announced between 1981 and 2000.
- The acquisition must be completed within 180 days of the announcement.
- The target must have at least one manufacturing SIC code, broadly SIC 2000 to 3999, in SDC or Compustat.
- The authors identify 2,030 SDC/Compustat manufacturing transactions and match 1,483 deals to the Longitudinal Research Database.

Interpretation: The sample is designed to study real operating assets, not only stock returns or accounting aggregates. The restriction to manufacturing is necessary because the plant-level Census data are manufacturing-based.

3.2 Plant-level Census/LRD data

- The Longitudinal Research Database tracks manufacturing plants over time.
- The Annual Survey of Manufactures provides plant-level information on output, employment, shipments, labor costs, materials, energy, and capital inputs.
- Larger plants are sampled with certainty, while smaller plants are sampled probabilistically.
- The authors record the owner of each target plant before the merger and follow ownership forward for three years after completion.
- A target plant is classified as *kept* if it remains with the acquirer, *sold* if it remains operating but is owned by another firm, and *closed* if it shuts down.

Interpretation: The key measurement advantage is asset tracking. Once a target firm is merged into an acquirer, ordinary firm-level data lose the target's internal assets. The Census plant identifiers allow the authors to keep following those assets.

3.3 Organizational form and relatedness

- The authors aggregate plants into firm-industry segments at the three-digit SIC level.
- A firm is classified as multi-segment if more than 10% of its sales are produced outside its principal three-digit SIC industry.
- A main segment is a segment whose shipments are at least 25% of total firm shipments.
- A target plant is related to the acquirer if it operates in the same three-digit SIC code as an acquirer’s main division.
- The variable *TMAIN* identifies whether a target plant belongs to a main division of the target.

Interpretation: Relatedness and main-division status are the paper’s main asset-fit measures. They allow the authors to distinguish plants that plausibly fit the acquirer’s capabilities from plants that may have been acquired as part of a larger bundle.

3.4 Productivity and operating margins

The paper uses two main plant-level performance measures.

- **Total factor productivity.** The authors estimate plant productivity from a translog production function using plant-level inputs and output.
- **Operating margin.** Plant operating margin equals shipments minus labor, materials, and energy costs, divided by shipments. It is industry-adjusted by subtracting the industry median.

The productivity regression is written conceptually as

$$\ln Q_{it} = A + f_i + \sum_{j=1}^N c_j \ln L_{jit} + \sum_{j=1}^N \sum_{k=j}^N c_{jk} \ln L_{jit} \ln L_{kit},$$

where Q_{it} is output of plant i in year t , L_{jit} is input j , and f_i is a plant-firm fixed effect.

Interpretation: The TFP measure captures how much output a plant produces relative to what would be predicted from its inputs. The margin measure captures operating profitability. Both focus on plant-level operating efficiency and exclude headquarters-level costs.

3.5 Industry, firm, and financial variables

- Industry controls include industry stock returns, R&D intensity, industry operating margin, and the dispersion of industry margins.
- Firm controls include acquirer output, number of acquirer plants, target size relative to acquirer size, and book-to-market ratios.
- *ASKILL* measures the acquirer’s skill in peripheral businesses, using productivity or margin in divisions outside the acquirer’s main divisions.
- *SYNERGY* is the product of acquirer and target industry-adjusted margins and is intended to capture complementarity.

- Financing variables include payment method, leverage, dividends, cash balances, long-term debt rating, and the Kaplan-Zingales financial-constraints index.

Interpretation: These variables let the paper separate asset-side explanations from financing-side explanations. The central claim is not that financing is irrelevant, but that asset fit and acquirer capability remain strong even after financing controls.

4 Models and empirical specifications

4.1 Theory: empire building versus comparative advantage

- **Empire-building view.** Managers like larger firms or overestimate their ability. This predicts excessive retention and weak systematic improvement of acquired plants.
- **Target-liberation view.** Acquisitions free assets trapped in inefficient targets. This predicts large gains especially for sold-off assets if new owners reallocate them efficiently.
- **Comparative-advantage view.** Firms keep assets they can operate well and dispose of assets with higher value elsewhere.
- **Marginal-skill prediction.** Acquirers with higher productivity in peripheral businesses should be more able to absorb additional plants and should retain more target assets.
- **Industry-shock prediction.** Positive industry shocks raise the opportunity cost of keeping assets, but high-skill acquirers should still be more likely to retain assets because they can operate them productively.

Interpretation: The paper’s most important theoretical move is to focus on marginal capability, not average firm quality. A firm can be highly profitable in its core but still poorly suited to operate additional peripheral plants.

4.2 Disposition model

The main disposition model is a multinomial logit with three outcomes: keep, sell, or close. Keeping is the baseline outcome. A generic version is

$$\Pr(Y_i = m) = \frac{\exp(X_i' \beta_m)}{1 + \sum_{k \in \{sell, close\}} \exp(X_i' \beta_k)}, \quad m \in \{sell, close\}.$$

The explanatory variables include target plant characteristics, acquirer characteristics, industry variables, and deal controls. The main variables are relatedness, target main-division status, target margin, acquirer margin, acquirer peripheral skill, industry returns, and the interaction between acquirer skill and industry returns.

Interpretation: The model asks whether plant disposition is predictable from economic fundamentals. A comparative-advantage view predicts more retention for related, central, productive, and acquirer-fit assets.

4.3 Financing-side specification

The paper adds payment method and financial-constraint variables to the disposition logits:

$$Y_i \in \{keep, sell, close\} = f(X_i, PCASH, KZ, Leverage, Dividend, Cash, Rating).$$

- *PCASH* identifies deals paid mostly with cash.
- *KZ* is a financial-constraints index.
- Leverage, dividends, cash balances, and credit ratings proxy for financial constraints and liquidity.

Interpretation: This specification asks whether acquirers sell plants because the assets do not fit, or because financial pressure forces sales. The answer is both, but asset-side variables remain central.

4.4 Post-merger performance regressions

The performance regressions estimate changes in productivity or margins from year $t - 1$ to year $t + 3$:

$$\Delta p_{i,-1,+3}^{adj} = \alpha + \beta_1 TMARG_i + \beta_2 AMARG_i + \beta_3 SYNERGY_i + \beta_4 ASKILL_i + \beta_5 TRELSIZE_i + \Gamma' Controls_i + \lambda \widehat{IMR}$$

- p is either TFP or operating margin.
- The dependent variable is adjusted for predictable time-series changes in performance.
- The inverse Mills ratio comes from a first-stage selection model for sell versus keep.
- The regressions are estimated separately for kept and sold plants.

Interpretation: This design asks whether kept plants actually improve after acquisition, and whether sold plants improve after being transferred again. It directly connects boundary choice to operating outcomes.

4.5 Additional agency tests

- The paper studies repeat acquirers, who might be especially likely to build empires.
- It studies business combination laws, which protect managers from takeover threats and could increase inefficient acquisitions.
- It also compares target plants with acquirers' own plants after the merger.

Interpretation: These tests do not rule out all agency problems. They ask a narrower question: does the real-asset side of post-merger restructuring look like pure empire building? The answer is generally no.

5 Identification and empirical design

5.1 What the design can identify

- The paper identifies systematic associations between plant characteristics, acquirer capabilities, industry conditions, and post-merger disposition.
- It also identifies whether plants kept by acquirers improve differently from plants sold by acquirers.
- The analysis is strongest as evidence about the allocation and performance of real assets after mergers.

Interpretation: The paper does not randomly assign target plants to acquirers. Its evidence is observational, but unusually detailed because the unit of analysis is the plant rather than the firm.

5.2 Benchmarks and counterfactuals

- The authors compare target-plant sales and closures with industry-year matched non-merger benchmarks.
- They compare whole-firm acquisitions with partial-firm asset purchases.
- They compare target plants with acquirers' own plants.
- Performance changes are adjusted for predictable mean reversion in plant performance.

Interpretation: These comparisons help show that the high disposal rates are merger-specific and not simply normal plant turnover.

5.3 Selection concerns

- Targets and acquirers are not randomly matched.
- Acquirers may choose targets partly because of unobserved fit.
- The sell-versus-keep decision is itself endogenous.
- The paper addresses selection in performance regressions using first-stage disposition models and an inverse Mills ratio.

Interpretation: The results should not be interpreted as a clean causal experiment. They are best read as showing that actual post-merger decisions line up closely with comparative-advantage predictions.

5.4 Measurement advantages and limitations

- The main advantage is direct tracking of plants before and after mergers.
- The sample is limited to manufacturing, so the conclusions may not fully generalize to services, software, or finance.

- Plant-level margins exclude headquarters overhead, advertising, and R&D costs.
- The Census matching process favors larger firms and plants.
- The paper cannot rule out overpayment, redistribution from acquirer to target shareholders, or agency problems outside plant operations.

Interpretation: The paper is a strong study of real operating assets, but it is not a complete welfare analysis of mergers.

6 Tables and figures

6.1 Figures 1–3: Comparative advantage and firm boundaries

Read:

- Figure 1 shows the basic boundary problem. The acquirer compares the marginal productivity of operating additional plants with the opportunity cost of selling them.
- The acquirer keeps plants up to the point where marginal productivity equals opportunity cost and sells the excess.
- Figure 2 shows a positive industry valuation shock that raises the acquirer's marginal product more than the opportunity cost. The acquirer keeps more plants.
- Figure 3 shows a positive shock that raises opportunity cost more than the acquirer's marginal product. The acquirer sells more plants.
- These diagrams motivate the interaction between acquirer skill and industry returns in the empirical logits.

Economic message: The figures translate the theory of firm boundaries into a testable empirical prediction: high-skill acquirers should keep more assets, especially when industry conditions make those assets valuable in productive hands.

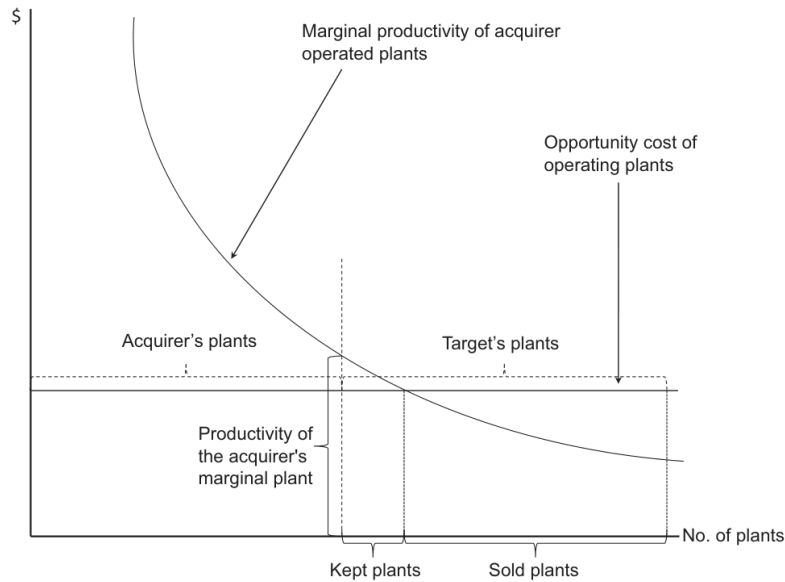
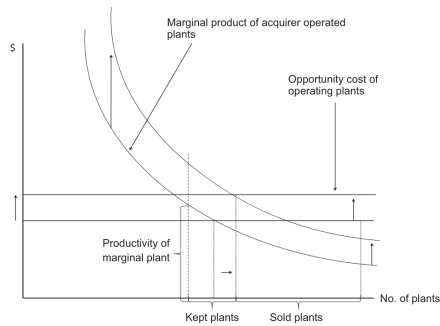
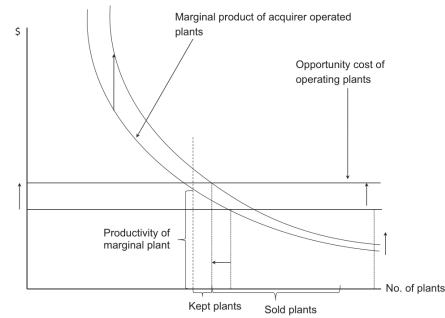


Figure 1: Marginal productivity, opportunity cost, and post-merger asset retention.



(a) Figure 2: Positive shock with a large productivity effect.



(b) Figure 3: Positive shock with a large opportunity-cost effect.

6.2 Tables 1 and 2: Sample construction and target/acquirer characteristics

Read:

- Table 1 reports the merger sample by year. The authors start with 2,030 SDC/Compustat manufacturing deals and match 1,483 to the LRD.
- Merger activity rises in the 1980s, declines in the early 1990s, and rises again in the late 1990s.
- Table 2 summarizes target and acquirer characteristics for the target-market-value sample.
- Targets tend to be small in market-cap deciles but not necessarily distressed.
- Acquirer plants generally have higher adjusted margins and larger deflated shipments than target plants.

Economic message: The sample is large enough to study plant-level restructuring over merger

waves. The target/acquirer comparison suggests that acquirers are typically larger and more productive operators.

Table 1

Number of deals.

The table describes the number of merger transactions in our study. The data comprise whole-firm acquisitions listed in the SDC M&A database in which the announcement date is between 1981 and 2000, the completion date is within 180 days of the announcement, and the acquisition target is a domestic U.S. firm with at least one reported four-digit SIC code from either SDC or Compustat between 2000 and 1999. Column 2 reports the number of transactions in the SDC Platinum that meet all criteria and match to Compustat. Column 3 reports the number of these transactions that were able to be matched to the Longitudinal Research Database maintained at the U.S. Department of Commerce. It includes 180 transactions that were coded as outside manufacturing by SDC and Compustat but were also found to have manufacturing assets.

Year	# Deals SDC/Compustat in manufacturing	Total # deals matched to LRD
1981	33	18
1982	59	46
1983	48	41
1984	70	58
1985	74	66
1986	125	104
1987	95	77
1988	154	115
1989	115	102
1990	62	59
1991	54	33
1992	44	28
1993	54	51
1994	79	48
1995	120	66
1996	115	93
1997	158	105
1998	186	113
1999	208	139
2000	177	121
Total	2,030	1,483

Table 2

Target and acquirer characteristics: target data market value available.

The table reports the mean and median (in parentheses) of selected characteristics of acquirers and targets. The sample consists of mergers from the SDC Platinum database in which the announcement date is between 1981 and 2000, the completion date is within 180 days of the announcement, the acquisition target is a domestic U.S. firm with at least one reported four-digit SIC code between 2000 and 1999, and the target has matching input/output data in the Longitudinal Research Database maintained at the U.S. Department of Commerce. The sample comprises firms for which the market value of the target is available. BE/ME decile and ME decile denote the book-to-market and NYSE market capitalization deciles to which the target belongs based on year $t-1$ values. The adjusted margin is the actual operating margin of a target plant minus the median margin for all plants that have the same three-digit SIC code. The deflated shipments equals the value of shipments for a plant reported in the ASM adjusted for inflation using the SIC deflator from the Bartelsman and Gray (1994) database.

Year	BE/ME decile	ME decile	Adjusted margin (%)		Deflated shipments	
	Target	Target	Acquirer	Target	Acquirer	Target
1981	5.61 (5)	3.89 (2.5)	1.28	0.87	254,814	152,447
1982	5.07 (4)	3.60 (3)	2.46	1.02	178,348	76,465
1983	4.65 (4)	2.5 (1)	3.62	0.57	81,614	42,277
1984	4.98 (5)	2.72 (1)	0.22	2.57	326,670	114,538
1985	4.69 (4)	3.32 (2)	1.69	0.83	237,487	154,729
1986	4.58 (4)	3.18 (2)	2.38	2.22	170,048	95,584
1987	4.85(4)	2.73 (2)	3.21	0.98	293,416	85,519
1988	5.08 (5)	2.63 (2)	5.64	4.36	195,577	119,498
1989	4.38 (4)	3.05 (2)	2.93	-0.48	135,143	65,729
1990	5.18 (4)	3.02 (2)	0.69	7.80	418,129	163,169
1991	5.84 (6)	2.94 (1.5)	0.92	-2.38	422,266	155,766
1992	3.75 (3)	2.93 (2)	1.57	1.57	430,009	55,630
1993	5.09 (4)	2.50 (2)	7.65	0.29	511,955	91,724
1994	4.72 (3)	2.94 (2)	5.63	3.65	377,213	94,126
1995	4.42 (4)	3.02 (2)	4.64	1.40	160,551	64,866
1996	4.76 (4)	2.65 (1)	3.72	2.96	322,041	111,002
1997	4.88 (4)	2.90 (2)	5.17	3.72	524,207	152,392
1998	5.39 (5)	3.06 (2)	3.82	3.07	1,225,467	154,534
1999	5.12 (5)	3.58 (3)	4.65	4.04	2,718,597	146,281
2000	4.73 (4)	3.49 (3)	4.91	3.58	1,455,530	107,710

(a) Table 1: Number of deals.

(b) Table 2: Target and acquirer characteristics.

6.3 Table 3: Disposition of target and acquirer plants

Read:

- Table 3 is the paper's first major empirical result.
- Across 12,893 target plants, only 54.42% are kept by the acquirer after three years.
- 18.58% of target plants are closed and 27.00% are sold.
- These rates are much larger than industry benchmarks: the matched-industry sales rate is 8.98%, and the matched-industry closure rate is 3.29%.
- Targeted partial-firm asset purchases have much lower subsequent sales and closures than whole-firm acquisitions.
- Related target plants are kept more often and sold less often than unrelated plants.

Economic message: Whole-firm mergers are followed by rapid boundary revision. Acquirers do not simply absorb targets; they actively sort target plants into kept, sold, and closed assets.

Table 3

Disposition of target and acquirer plants.

The table reports the year +3 ownership status of plants, where the merger is completed in year 0. The sample consists of mergers from the SDC Platinum database in which the announcement date is between 1981 and 2000, the completion date is within 180 days of the announcement, the acquisition target is a domestic U.S. firm with at least one reported four-digit SIC code between 2000 and 3999, and the target has matching input/output data in the Longitudinal Research Database maintained at the U.S. Department of Commerce. Kept plants are still owned by the acquirer, sold plants are owned by a firm other than the acquirer, and closed plants are plants that shut down as of year +3. In each period, we classify the deals by the number of target plants acquired for target disposition and by the number of acquirer plants for acquirer disposition. 1980s transactions have a completion date between 1981 and 1989 and 1990s transactions form the complementary set. Industry benchmarks for asset sales and closures are from industries that experience a merger transaction in the same three-digit SIC code and year. A target plant is related if it belongs to the same three-digit SIC code as a main division of the acquirer.

(1) # Plants in deal	(2) # Total bought	(3) % Kept target	(4) % Closed target	(5) % Closed matched industry	(6) % Sold target (partial)	(7) % Sold target (total)	(8) % Sold matched industry	(9) % Sold acquirer (total)	(10) % Closed acquirer
<i>Panel A: Full sample</i>									
1–5	1,954	56.59%	16.15%	2.40%	9.29%	27.20%	7.19%	14.17%	3.13%
6–10	1,193	53.76%	21.38%	4.23%	13.67%	24.86%	10.8%	12.51%	3.88%
11–25	2,316	54.69%	19.96%	4.57%	17.79%	25.35%	12.62%	14.12%	5.25%
26–50	3,337	56.57%	16.98%	5.19%	22.07%	26.45%	14.01%	18.49%	5.02%
≥ 51	4,093	51.76%	19.44%	5.18%	26.07%	28.80%	10.96%	16.08%	5.08%
Total	12,893	54.42%	18.58%	3.29%	19.99%	27.00%	8.98%	14.69%	4.02%
<i>Panel B: By time period</i>									
1980s	6,710	50.33%	19.42%	3.71%	23.40%	30.25%	10.81%	15.57%	3.88%
1990s	6,183	59.15%	17.61%	2.89%	16.04%	23.24%	7.33%	13.95%	4.14%
<i>Panel C: By relatedness</i>									
Related	4,080	54.78%	17.72%		14.12%	22.53%			
Unrelated	8,813	51.02%	17.87%		21.51%	27.46%			

Table 3: Disposition of target and acquirer plants.

6.4 Tables 4 and 5: Multinomial logit for plant disposal

Read:

- Table 4 reports multinomial logit coefficients for selling or closing target plants relative to keeping them.
- Table 5 reports marginal effects for keeping, selling, and closing.
- Related plants and target-main-division plants are significantly less likely to be sold or closed.
- Higher target plant margins reduce sale and closure probabilities.
- Acquirer peripheral skill increases retention. A one-standard-deviation increase in acquirer marginal skill raises retention probabilities by roughly 17–19% in the specifications without the interaction.
- The interaction between acquirer skill and industry returns supports the comparative-advantage prediction: skilled acquirers are less likely to sell plants after positive industry shocks.
- The 1980s dummy is associated with more disposal and less retention, consistent with the view that 1980s mergers involved more deconglomeration.

Economic message: The disposal decision follows economically sensible patterns. Acquirers keep assets that fit their operations and sell assets that are peripheral, unrelated, low-margin, or especially valuable to other owners.

Table 4

Multinomial logit for plant disposal.

The table reports logit estimates of the decision to sell or close acquired plants by year + 3 after acquisition completion for the sample used in Table 3. RELATED is 1 if the target's main business overlaps with an acquirer main division and zero otherwise. TMAIN is 1 if the plant's output is at least 25% of the total target output and zero otherwise. TBEME is the target's book-to-market ratio decile, AMARG (TMARG) denote the acquirer (target) operating margin at the firm (plant) level less the three-digit SIC median. ASKILL is the average three-digit SIC industry-adjusted margin of acquirer plants outside the main divisions. IND R&D (INDMARG) denotes the aggregate R & D expenditure (operating margin) of all firms in the plant's three-digit SIC code. INDRRET is the $(t, t+2)$ buy-and-hold return for the plant's Fama-French 48-industry group. SYNERGY is the product of acquirer and target industry-adjusted margins. SD(INDMARG) is the standard deviation of operating margins of all plants in the same three-digit SIC code. TRELSIZE is the ratio of the aggregate deflated output of the target to that of the acquirer. ANUMPLANT and LN(AOUTPUT) denote the number of plants owned by and log deflated output of the acquirer. 1980s is 1 for mergers completed between 1981 and 1989 and zero otherwise. z -statistics based on robust standard errors, clustered at the industry level with correlation within firm panel units, are reported in parentheses.

Independent variable	Specification #				
	(1)	(2)	(3)	(4)	(5)
<i>Panel A: Dependent variable: decision to sell plant</i>					
RELATED	-0.80 (-8.73) ^a	-0.81 (-8.67) ^a	-0.75 (-7.93) ^a	-0.73 (-7.63) ^a	-0.72 (-7.59) ^a
TMAIN	-0.91 (-13.59) ^a	-0.91 (-13.69) ^a	-0.95 (-14.02) ^a	-0.95 (-13.94) ^a	-0.96 (-13.88)
TBEME	0.09 (1.77) ^a	0.08 (1.62)	0.10 (2.03) ^a	0.09 (1.81) ^a	0.09 (1.70) ^a
TMARG	-0.76 (-6.42) ^a	-0.75 (-6.22) ^a	-0.78 (-6.36) ^a	-0.78 (-6.4) ^a	-0.73 (-5.29) ^a
AMARG			-0.31 (-0.98)	-0.31 (-1.00)	-0.35 (-1.11)
INDRET			0.19 (2.37) ^a	0.20 (2.54) ^a	0.22 (2.84) ^a
ASKILL		-0.94 (-2.47) ^b	-1.00 (-2.59) ^b	-0.093 (-0.19)	-0.033 (-0.08)
ASKILL* INDRRET				-3.36 (-3.28) ^a	-3.22 (-3.05) ^a
SYNERGY					-4.11 (-3.87) ^a
IND R&D			3.69 (2.83) ^a	3.53 (2.7) ^a	3.56 (2.70)
SD (INDMARG)			-0.55 (-0.56)	-0.563 (-0.57)	-0.34 (-0.35)
INDMARG			1.01 (3.23) ^a	0.99 (3.18) ^a	1.01 (3.24) ^a
TRELSIZE	0.08 (1.54)	0.08 (1.52)	0.06 (1.24)	0.06 (1.26)	0.06 (1.17)
LN (AOUTPUT)	0.06 (2.45) ^b	0.05 (2.37) ^b	0.05 (2.16) ^b	0.05 (2.18) ^b	0.04 (1.88) ^a
ANUMPLANT	0.00 (0.66)	0.00 (0.67)	0.00 (0.80)	0.00 (0.73)	0.00 (0.75)
1980s	0.28 (4.54) ^a	0.27 (4.45) ^a	0.32 (5.07) ^a	0.32 (4.98) ^a	0.31 (4.76) ^a
Constant	-0.86 (-2.79) ^a	-0.82 (-2.63) ^a	-1.06 (-2.91) ^a	-1.07 (-2.93) ^a	-1.01 (-2.73) ^a
<i>Panel B: Dependent variable: decision to close plant</i>					
RELATED	-0.43 (-4.24) ^a	-0.43 (-4.18) ^a	-0.35 (-3.33) ^a	-0.35 (-3.36) ^a	-0.35 (-3.33)
TMAIN	-0.46 (-6.72) ^a	-0.46 (-6.77) ^a	-0.49 (-7.01) ^a	-0.49 (-7.02) ^a	-0.48 (-6.98)
TBEME	-0.01 (-0.22)	-0.01 (-0.22)	0.05 (0.86)	0.04 (0.85)	0.05 (0.93)
TMARG	-0.79 (-6.17) ^a	-0.79 (-6.17) ^a	-0.75 (-5.87) ^a	-0.75 (-5.86) ^a	-0.64 (-4.07)
AMARG		0.12 (0.39)	-0.14 (-0.46)	-0.15 (-0.49)	-0.15 (-0.46)
INDRET			0.06 (0.67)	0.06 (0.70)	0.08 (0.96)
ASKILL		-0.45 (-1.17)	-0.54 (-1.40)	-0.64 (-1.42)	-0.67 (-1.57)
ASKILL* INDRRET				0.19 (0.19)	-0.07 (-0.08)
SYNERGY					-3.06 (-2.61) ^a
IND R&D			4.86 (3.72) ^a	4.88 (3.73) ^a	4.94 (3.73) ^a
SD (INDMARG)			2.55 (2.40) ^b	2.55 (2.4) ^b	2.80 (2.40) ^b
INDMARG			1.10 (3.37) ^a	1.09 (3.33) ^a	1.13 (3.26) ^a
TRELSIZE	-0.12 (-2.36) ^b	-0.12 (-2.38) ^b	-0.12 (-2.33) ^b	-0.12 (-2.3) ^b	-0.11 (-2.04) ^b
LN (AOUTPUT)	-0.14 (-5.69) ^a	-0.14 (-5.76) ^a	-0.13 (-5.42) ^a	-0.13 (-5.4) ^a	-0.13 (-5.22) ^a
ANUMPLANT	0.00 (0.66)	0.00 (0.72)	0.00 (1.23)	0.00 (1.22)	0.00 (1.14)
1980s	0.35 (5.39) ^a	0.34 (5.36) ^a	0.43 (6.44) ^a	0.43 (6.45) ^a	0.44 (6.45) ^a
Constant	1.21 (3.82) ^a	1.23 (3.88) ^a	0.16 (0.42)	0.17 (0.45)	0.12 (0.30)
N	8,164	8,164	8,026	8,026	7,859
Pseudo-R ²	0.04	0.04	0.045	0.046	0.048

^a = significant at 1%, ^b = significant at 5%, ^c = significant at 10%.

Standard errors are robust and control for industry and firm clustering. Table 4 presents the logit coefficients and Table 5 presents marginal effects including the implied

interactions with industry variables, which allow us to further test predictions about efficient disposal decisions. Specification (1) reports the effect of the target plant

Table 5

Logit marginal effects.

The table reports marginal effects for the logit estimates in Table 4. RELATED is 1 if the target's main business overlaps with an acquirer main division and zero otherwise. TMAIN is 1 if the plant's output is at least 25% of the total target output and zero otherwise. TBEME is the target's book-to-market ratio decile. AMARG (TMARG) denote the acquirer (target) operating margin at the firm (plant) level less the three-digit SIC median. ASKILL is the average three-digit SIC industry-adjusted margin of acquirer plants outside the main divisions. IND R&D (INDMARG) denotes the aggregate R & D expenditure (operating margin) of all firms in the plant's three-digit SIC code. INDRRET is the $(t, t+2)$ buy-and-hold return for the plant's Fama-French 48-industry group. SYNERGY is the product of acquirer and target industry-adjusted margins. SD(INDMARG) is the standard deviation of operating margins of all plants in the same three-digit SIC code. TRELSIZE is the ratio of the aggregate deflated output of the target to that of the acquirer. ANUMPLANT and LN(AOUTPUT) denote the number of plants owned by and log deflated output of the acquirer. 1980s is 1 for mergers completed between 1981 and 1989 and zero otherwise. z -statistics based on robust standard errors, clustered at the industry level with correlation within firm panel units, are reported in parentheses.

Independent variable	(1)	(2)	Specification # (3)	(4)	(5)
<i>Marginal effect on keep decision</i>					
RELATED	0.16 (8.02) ^a	0.16 (7.96) ^a	0.14 (7.00) ^a	0.14 (6.81) ^a	0.14 (6.73) ^a
TMAIN	0.16 (13.48) ^a	0.17 (13.58) ^a	0.17 (13.99) ^a	0.17 (13.94) ^a	0.17 (13.67) ^a
TBEME	-0.01 (-1.00)	-0.01 (-0.90)	-0.18 (-1.83) ^a	-0.02 (-1.67) ^a	-0.02 (-1.71) ^a
TMARG	0.18 (7.87) ^a	0.18 (7.66) ^a	0.18 (7.56) ^a	0.17 (7.57) ^a	0.17 (5.88) ^a
AMARG		0.00 (-0.07)	0.05 (0.93)	0.06 (0.95)	0.06 (0.99)
ASKILL		0.17 (2.3) ^a	0.19 (2.52) ^a	0.08 (0.92)	0.08 (0.94)
INDRET			-0.03 (-1.94) ^a	-0.03 (-2.07) ^a	-0.04 (-2.42) ^a
ASKILL* INDRRET				0.39 (2.02) ^a	0.44 (2.26) ^a
SYNERGY					0.87 (4.01) ^a
IND R&D			-1.02 (-3.98) ^a	-1.00 (-3.91) ^a	-1.00 (-3.95) ^a
SD (INDMARG)			-0.21 (-1.07)	-0.21 (-1.07)	-0.27 (-1.32)
INDMARG			-0.25 (-4.09) ^a	-0.25 (-4.03) ^a	-0.25 (-4.05) ^a
TRELSIZE	0.00 (0.34)	0.00 (0.37)	0.00 (0.49)	0.00 (0.49)	0.00 (0.44)
LN (AOUTPUT)	0.01 (1.82) ^a	0.01 (1.92) ^a	0.01 (1.80) ^a	0.01 (1.80) ^a	0.01 (1.91) ^a
ANUMPLANT	0.00 (-0.82)	0.00 (-0.86)	0.00 (-1.25)	0.00 (-1.28)	0.00 (-1.19)
1980s	-0.07 (-6.28) ^a	-0.07 (-6.2) ^a	-0.09 (-7.26) ^a	-0.09 (-7.24) ^a	-0.09 (-7.03) ^a
<i>Marginal effect on sell decision</i>					
RELATED	-0.13 (-7.22) ^a	-0.13 (-7.17) ^a	-0.13 (-6.74) ^a	-0.12 (-6.46) ^a	-0.12 (-6.42) ^a
TMAIN	-0.13 (-13.03) ^a	-0.13 (-13.13) ^a	-0.14 (-13.46) ^a	-0.14 (-13.37) ^a	-0.14 (-13.45)
TBEME	0.01 (1.91) ^a	0.01 (1.75) ^a	0.02 (1.90) ^a	0.01 (1.67) ^a	0.01 (1.55)
TMARG	-0.10 (-5.08) ^a	-0.09 (-4.89) ^a	-0.10 (-5.17) ^a	-0.10 (-5.21) ^a	-0.10 (-4.48) ^a
AMARG		-0.02 (-0.34)	-0.05 (-0.91)	-0.05 (-0.93)	-0.05 (-1.04)
ASKILL		-0.14 (-2.3) ^a	-0.15 (-2.36) ^a	0.01 (0.12)	0.02 (0.31)
INDRET			0.03 (2.3) ^a	0.03 (2.47) ^a	0.03 (2.74) ^a
ASKILL* INDRRET				-0.55 (-3.21) ^a	-0.56 (-3.48) ^a
SYNERGY				-0.57 (-3.37) ^a	-0.57 (-3.37) ^a
IND R&D			0.43 (2.05) ^a	0.40 (1.92) ^a	0.40 (1.91) ^a
SD (INDMARG)			-0.19 (-1.21)	-0.20 (-1.22)	-0.17 (-1.05)
INDMARG			0.13 (2.51) ^a	0.12 (2.46) ^a	0.12 (2.50) ^a
TRELSIZE	0.02 (2.19) ^b	0.02 (2.17) ^b	0.02 (1.85) ^a	0.02 (1.88) ^a	0.01 (1.75) ^a
LN (AOUTPUT)	0.01 (4) ^a	0.01 (3.93) ^a	0.01 (3.61) ^a	0.01 (3.63) ^a	0.01 (3.60) ^a
ANUMPLANT	0.00 (0.50)	0.00 (0.50)	0.00 (0.50)	0.00 (0.44)	0.00 (0.47)
1980s	0.03 (3.34) ^a	0.03 (3.25) ^a	0.04 (3.65) ^a	0.04 (3.55) ^a	0.03 (3.36) ^a
<i>Marginal effect on close decision</i>					
RELATED	-0.03 (-1.68) ^a	-0.03 (-1.62)	-0.02 (-1.04)	-0.02 (-1.17)	-0.02 (-1.20)
TMAIN	-0.03 (-3.46) ^a	-0.03 (-3.48) ^a	-0.04 (-3.69) ^a	-0.04 (-3.73) ^a	-0.04 (-3.49) ^a
TBEME	-0.01 (-0.67)	0.00 (-0.64)	0.00 (0.38)	0.00 (0.42)	0.00 (0.47)
TMARG	-0.09 (-4.88) ^a	-0.09 (-4.85) ^a	-0.08 (-4.50) ^a	-0.08 (-4.48) ^a	-0.07 (-2.99) ^a
AMARG		0.02 (0.47)	-0.01 (-0.20)	-0.01 (-0.22)	-0.01 (-0.19)
ASKILL		-0.03 (-0.54)	-0.04 (-0.76)	-0.09 (-1.40)	-0.10 (-1.64) ^a
INDRET			0.00 (0.08)	0.00 (0.07)	0.00 (0.24)
ASKILL* INDRRET				0.16 (1.11)	0.12 (0.97)
SYNERGY					-0.30 (-1.78) ^a
IND R&D			0.39 (3.14) ^a	0.60 (6.17) ^a	0.60 (6.17) ^a
SD (INDMARG)			0.41 (2.63) ^a	0.41 (2.65) ^a	0.43 (2.57) ^a
INDMARG			0.13 (2.68) ^a	0.13 (2.62) ^a	0.13 (2.62) ^a
TRELSIZE	-0.02 (-2.89) ^a	-0.02 (-2.91) ^a	-0.20 (-2.78) ^a	-0.02 (-2.80) ^a	-0.02 (-2.44) ^b
LN (AOUTPUT)	-0.02 (-6.57) ^a	-0.02 (-6.61) ^a	-0.02 (-6.21) ^a	-0.02 (-6.23) ^a	-0.02 (-5.92) ^a
ANUMPLANT	0.00 (0.52)	0.00 (0.58)	0.00 (1.08)	0.00 (1.09)	0.00 (1.00)
1980s	0.04 (4.47) ^a	0.04 (4.45) ^a	0.05 (5.45) ^a	0.05 (5.48) ^a	0.05 (5.58) ^a

^a = significant at 1%, ^b = significant at 5%, ^c = significant at 10%.

(a) Table 4: Multinomial logit for plant disposal.

(b) Table 5: Logit marginal effects.

6.5 Tables 6 and 7: Financing variables and unconditional performance changes

Read:

- Table 6 adds method of payment and financial-constraint variables to the plant-disposition logits.
- Cash payment weakly reduces the likelihood of plant sales, especially when cash comes from internal resources.
- Financial constraints have stronger effects: higher leverage and higher KZ are associated with more sales, while dividends and cash balances are associated with fewer sales.
- Table 7 shows unconditional post-acquisition performance changes for target plants.
- Kept target plants improve substantially: by year $t + 3$, TFP rises about 6.3% and margins rise about 2.1 percentage points in the full sample.
- Sold plants improve much less and often only weakly.

Economic message: Financing conditions affect restructuring, but they do not replace the asset-side story. The performance evidence is especially important: acquirers keep the plants that improve most under their ownership.

Table 6
Multinomial logit—method of payment, leverage, and financing.

The table reports estimates of a multinomial logit that models an acquirer's decision to sell or close a target plant relative to the baseline of keeping the plant. PCASH is 1 if cash is used to pay for at least 51% of the acquisition cost. PCASH INTERNAL is 1 if the acquirer uses its own cash resources to pay for the deal and zero otherwise. PCASH DEBT is 1 if debt is used to finance part of the cash payment and zero otherwise. KZ is the Kaplan-Zingales (1997) financial constraints index. LEVERAGE is the acquirer's debt-to-assets ratio DIV is 1 if the acquirer pays dividends and is zero otherwise. CASH denotes the acquirer's cash balance to its book value of assets. LT RATING is 1 if the acquirer has a credit rating and is zero otherwise. The models include controls used in the full specification #6 of Table 4. Robust t -statistics that allow for industry clustering and correlation within firm panels are in parentheses.

Independent variable	Specification #					
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Dependent variable: sell plant</i>						
PCASH	-0.167 (-1.65) [*]	-0.155 (-1.52)	-0.157 (-1.51)			
PCASH INTERNAL				-0.274 (-2.28) [*]	-0.272 (-2.26) [*]	-0.253 (-1.97) [*]
PCASH DEBT				-0.077 (-0.66)	-0.054 (-0.46)	-0.083 (-0.72)
KZ		0.012 (3.88) [*]			0.013 (3.94) [*]	
LEVERAGE			1.389 (4.04) [*]			1.414 (4.09) [*]
DIV			-0.398 (-3.15) [*]			-0.406 (-3.20) [*]
CASH			-4.173 (-4.30) [*]			-4.041 (-4.12) [*]
LT RATING			-0.088 (-0.69)			-0.102 (-0.80)
CONTROLS FROM Table 4	Yes	Yes	Yes	Yes	Yes	Yes
<i>Dependent variable: close plant</i>						
PCASH	0.001 (0.01)	0.017 (0.16)	0.028 (0.26)			
PCASH INTERNAL				-0.038 (-0.30)	-0.032 (-0.25)	-0.006 (-0.04)
PCASH DEBT				0.035 (0.29)	0.061 (0.50)	0.056 (0.46)
KZ		0.009 (2.29) [*]			0.009 (2.33) [*]	
LEVERAGE			0.695 (2.40) [*]			0.698 (2.41) [*]
DIV			-0.314 (-2.13) [*]			-0.318 (-2.15) [*]
CASH			-0.937 (-1.31)			-0.898 (-1.25)
LT RATING			0.057 (0.31)			0.031 (0.26)
CONTROLS FROM Table 4	Yes	Yes	Yes	Yes	Yes	Yes
Pseudo-R ²	0.091	0.093	0.101	0.091	0.094	0.101
N	3,757	3,757	3,757	3,757	3,757	3,757

Table 7

Post-acquisition changes in performance.

The table reports the average total factor productivity (TFP) and excess operating margin in year -1 and the changes in TFP and excess operating margin between year -1 and years $+1$, $+2$, and $+3$ where a full-firm merger is completed in year 0. Acquired plants are classified as Kept if the acquirer retains ownership of plants as of year $+3$ and as Sold if he plant was operating but not owned by the acquirer as of year $+3$. We report statistics for two efficiency measures: (1) Operating margin, which is ratio of the operating income before depreciation to the total plant shipments minus the industry median margin; (2) TFP, which is a plant's log output minus the predicted output based on a log-linear production function with squared and cross-product terms estimated for all plants in the industry. The sample consists of mergers from the SDC M&A database announced between 1981 and 2000 and completed within 180 days of announcement, in which the target is a domestic U.S. firm with at least one reported four-digit SIC code between 2000 and 3999 and has matching input/output data in the Longitudinal Research Database maintained at the U.S. Department of Commerce. We report two sets of estimates, one for all target plants and one for all target plants for which the acquirer's book-to-market is available in Compustat. Standard errors are in parentheses and the number of plants are below each performance statistic.

Statistic	Full sample				Sample with acquirer BE/ME			
	π_{-1}	$\Delta\pi_{-1,+1}$	$\Delta\pi_{-1,+2}$	$\Delta\pi_{-1,+3}$	π_{-1}	$\Delta\pi_{-1,+1}$	$\Delta\pi_{-1,+2}$	$\Delta\pi_{-1,+3}$
<i>Kept plants</i>								
$\pi = \text{TFP}$	0.201 (19.70) [*]	0.063 (7.56) [*]	0.081 (8.91) [*]	0.063 (6.61) [*]	0.21 (17.64) [*]	0.057 (5.76) [*]	0.094 (8.87) [*]	0.064 (5.31) [*]
N	6,348	6,346	6,346	6,346	4,452	4,452	4,452	4,452
$\pi = \text{Margin}$	0.032 (12.22) [*]	0.011 (5.34) [*]	0.011 (5.10) [*]	0.021 (9.24) [*]	0.036 (11.42) [*]	0.017 (6.75) [*]	0.012 (4.55) [*]	0.022 (7.94) [*]
N	6,409	6,409	6,409	6,409	4,452	4,452	4,452	4,452
<i>Sold plants</i>								
$\pi = \text{TFP}$	0.047 (3.28) [*]	0.013 (1.05)	0.022 (1.60)	0.027 (1.87) [*]	0.055 (2.85) [*]	0.006 (0.34)	0.016 (0.87)	0.027 (1.45)
N	2,871	2,871	2,871	2,871	1,530	1,530	1,530	1,530
$\pi = \text{Margin}$	0.002 (0.63)	-0.001 (-0.37)	0.003 (0.75)	0.007 (1.95) [*]	-0.007 (-1.38)	0.002 (0.49)	-0.003 (-0.57)	-0.003 (-0.54)
N	2,905	2,905	2,905	2,905	1,530	1,530	1,530	1,530

t -statistics from test of significance of the average from zero in parentheses.

* = significant at 1%; ** = significant at 5%; *** = significant at 10%.

When we separate the acquired plants into those sold 6.2. Changes in performance and acquirer and target

(a) Table 6: Method of payment, leverage, and financing.
in performance.

(b) Table 7: Post-acquisition changes

6.6 Tables 8 and 9: Performance regressions and financing-side variables

Read:

- Table 8 estimates cross-sectional regressions of post-acquisition performance changes for kept and sold plants.
- For kept plants, acquirer margin and synergy are positive predictors of performance improvement.
- Target margin enters negatively, meaning initially weaker plants have more room to improve after being retained.
- Larger targets relative to acquirers have lower TFP gains, consistent with decreasing returns to managerial scope.

- Sold plants show fewer robust performance patterns.
- Table 9 adds financing variables to the performance regressions.
- Payment method has little direct relation to productivity or margin improvements.
- Liquidity and financial-constraint variables have some relation to performance, especially operating margins of kept plants.

Economic message: The retained-plant performance results support the comparative-advantage story. The acquirer improves plants where its operating capabilities and complementarities are useful; sold plants do not show the same gains.

Table 8

Post-acquisition changes in performance: regressions.

The table reports regression estimates. The dependent variable is either the change in the total factor productivity (TFP) or the change in the excess operating margin between years -1 and $+3$ where the acquisition is completed in year 0. The sample comprises all acquired plants that were either kept (the first two columns of results) or sold (the last two columns of results). TMARG (AMARG) is the industry-adjusted target (acquirer) operating margin while ASKILL is the average of this margin over acquirer plants outside its main division. SYNERGY denotes the product of the acquirer and target operating margins. TRELSIZE is the acquirer-to-target ratio of aggregate deflated output. TBEME is the decile to which a target's book-to-market ratio belongs. 1980s is 1 if the merger is completed between 1981 and 1989 and zero otherwise. ABEME is the decile to which the acquirer's book-to-market ratio belongs. λ is the inverse Mills ratio from a probit model (estimates not reported to conserve space) in which the dependent variable is 1 if a plant is sold and zero if a plant is kept and the independent variables are as in Table 4. Z-statistics reported in parentheses are based on robust standard errors allowing for clustering at the industry level and correlation within firm panel units.

	Kept plants		Sold plants	
Panel A: Dependent variable = Δ TFP				
TMARG	-0.145 (-2.41) ^a	-0.280 (-3.48) ^a	-0.35 (-3.29) ^a	-0.304 (-1.73) ^c
AMARG	0.541 (4.81) ^a	0.611 (3.68) ^a	0.011 (0.05)	0.128 (0.41)
SYNERGY	3.101 (7.00) ^a	3.517 (5.64) ^a	0.483 (0.53)	1.335 (1.01)
ASKILL	-0.257 (-0.92)	-0.179 (-1.16)	-0.110 (-0.39)	-0.214 (-0.68)
TRELSIZE	-0.024 (-2.66) ^a	-0.056 (-4.45) ^a	-0.034 (-2.02)	-0.068 (-2.37)
TBEME	-0.044 (-2.19) ^b	-0.012 (-0.34)	0.004 (0.14)	0.076 (1.24)
1980s	-0.023 (-0.96)	-0.077 (-2.19)	-0.000 (-0.01)	-0.087 (-1.11)
ABEME		-0.046 (-0.93)		0.168 (1.43)
λ	-0.104 (-1.44)	-0.148 (-1.56)	0.020 (0.25)	-0.059 (-0.50)
CONSTANT	-0.069 (-1.62)	-0.168 (-2.67) ^b	-0.126 (-1.01)	-0.273 (-1.36)
N	4,238	2,356	1,452	671
Adjusted R ²	0.023	0.033	0.006	0.011
Panel B: Dependent variable = Δ excess operating margin				
TMARG	-0.026 (-1.50)	-0.053 (-2.24) ^b	-0.139 (-4.72) ^a	-0.159 (-3.48) ^a
AMARG	0.541 (4.81) ^a	0.611 (3.68) ^a	0.011 (0.05)	0.128 (0.41)
SYNERGY	0.363 (3.26) ^a	0.396 (2.20) ^b	0.028 (0.11)	0.322 (1.09)
ASKILL	-0.029 (-0.90)	-0.020 (-0.53)	-0.046 (-0.55)	-0.082 (-0.93)
TRELSIZE	-0.002 (-0.74)	-0.005 (-1.66)	-0.011 (-2.39) ^b	-0.013 (-1.55)
TBEME	0.004 (0.89)	0.015 (1.75) ^c	-0.002 (-0.27)	0.015 (0.94)
1980s	-0.014 (-0.93)	-0.017 (-1.87) ^c	0.012 (1.06)	0.025 (1.28)
ABEME		-0.038 (-2.94) ^a		-0.022 (-0.78)
λ	-0.043 (-2.43) ^b	-0.032 (-1.36)	0.006 (0.26)	0.001 (0.03)
CONSTANT	-0.021 (-1.94)	-0.016 (-0.98)	-0.058 (-1.64)	-0.074 (-1.26)
N	4,452	2,475	1,530	707
Adjusted R ²	0.017	0.033	0.022	0.028

^a = significant at 1%, ^b = significant at 5%, ^c = significant at 10%.

Table 9

Post-acquisition change in performance: financing-side variables.

The table reports regression estimates. The dependent variable is either the change in the total factor productivity (TFP) or the change in the excess operating margin between years -1 and $+3$ where the acquisition is completed in year 0. The sample comprises all acquired plants that were either kept (the first two columns of results) or sold (the last two columns of results). The explanatory variables include as controls all variables in Table 8 and additional financing-side variables. PCASH is 1 if cash is used to pay for at least 51% of the acquisition cost. LEVERAGE is the acquirer's debt-to-assets ratio. DIV is 1 if the acquirer pays dividends and is zero otherwise. CASH denotes the acquirer's cash balance to its book value of assets. LT RATING is 1 if the acquirer has a credit rating and is zero otherwise. All models control for the variables used in Table 8. Robust Z-statistics that allow for industry clustering and correlation within firm panels are in parentheses.

<i>Panel A: Dependent variable = Δ TFP</i>					
Independent variables	Kept plants		Sold plants		
PCASH	0.017 (0.29)	0.041 (0.69)	-0.054 (-0.99)	-0.051 (-0.83)	
LEVERAGE		-0.128 (-0.94)		-0.089 (-0.35)	
DIV		0.111 (1.81) ^c		0.019 (0.25)	
CASH		0.099 (0.25)		1.066 (1.37)	
LT RATING		0.090 (1.44)		0.036 (0.40)	
TABLE 8 CONTROLS	Yes	Yes	Yes	Yes	
Adjusted R ²	0.038	0.041	0.051	0.058	
N	2,049	2,049	623	623	
<i>Panel B: Dependent variable = Δ excess operating margin</i>					
Independent variables	Kept plants		Sold plants		
PCASH	-0.015 (-0.91)	-0.014 (-0.80)	0.015 (0.93)	0.023 (1.62)	
LEVERAGE		-0.080 (-3.14) ^a		-0.049 (-0.93)	
DIV		0.041 (3.77) ^a		0.002 (0.15)	
CASH		1.066 (1.37)		0.369 (1.79)	
LT RATING		0.023 (2.07) ^b		0.022 (0.92)	
TABLE 8 CONTROLS	Yes	Yes	Yes	Yes	
Adjusted R ²	0.026	0.023	0.047	0.049	
N	2,049	2,049	623	623	

^a = significant at 1%, ^b = significant at 5%, ^c = significant at 10%.

(a) Table 8: Post-acquisition changes in performance, regressions.

(b) Table 9: Performance regressions with financing-side variables.

6.7 Tables 10 and 11: Repeat acquirers and acquirer's own plants

Read:

- Table 10 examines repeat acquirers.
- Repeat acquirers do not keep more target plants. Plants in third or later deals are more likely to be sold.
- Performance results for repeat acquirers do not show destructive empire building. Fourth-and-later acquisitions are associated with stronger improvements for kept plants.
- Table 11 studies plants that the acquirer already owned before the acquisition.
- Acquirers' own kept plants also improve: TFP rises about 6.9% over three years in the full sample, and margins rise about 1.0 percentage point.
- Sold acquirer plants display weaker or mixed performance changes.

Economic message: Repeat-acquirer evidence does not support the idea that serial acquirers are simply empire builders. The acquirer's own plant results show that post-merger restructuring affects the whole firm, not only target assets.

Table 10

Repeat acquirers: disposition and performance changes.

Panel A reports estimates of a multinomial logit specification that models an acquirer's decision to sell or close a target plant relative to the baseline of keeping the plant. DEANUM2, DEANUM3, and DEANUM4+ are dummy variables that equal 1 if the acquisition is the second, third, or at least the fourth acquisition completed by an acquirer in our sample, and zero otherwise. The models include controls used in the full specifications in Tables 4. Panel B reports estimates in which the dependent variable is either the change in the total factor productivity (TFP) or the change in operating margin between year -1 and year $+3$, for a plant owned by the acquirer prior to the merger. Robust Z-statistics that allow for industry clustering and correlation within firm panels are in parentheses. a, b, and c denote significance at the 1%, 5%, and 10% levels.

Panel A: Logit estimates: target plant disposition				
Independent variable	Decision to			
	Sell plant	Close plant		
	Logit coefficients			
DEALNUM2	0.07 (0.71)	0.01 (0.83)		
DEALNUM3	0.55 (4.64) ^a	0.10 (4.26) ^a		
DEALNUM4 +	0.55 (4.45) ^a	0.11 (4.26) ^a		
Table 4 controls	Yes	Yes		
N	8,026	8,026		
Panel B: Target plant performance				
Independent variable	π = TFP		π = Operating margin	
	Performance of kept plants			
DEALNUM2	0.01 (0.23)	0.07 (1.56)	0.01 (1.34)	0.03 (3.02)
DEALNUM3	-0.12 (-2.17) ^b	-0.10 (-1.61) ^a	-0.02 (-1.35)	-0.02 (-1.56)
DEALNUM4 +	0.15 (3.14) ^a	0.26 (3.74) ^a	0.03 (2.37) ^b	0.04 (2.61) ^b
Table 8 controls	Yes	Yes	Yes	Yes
F-statistic	6.83 (0.00)	7.50 (0.00)	112.62 (0.00)	59.74 (0.00)
N	4,239	2,356	4,452	2,475
	Performance of sold plants			
DEALNUM2	0.048 (0.77)	-0.05 (-0.53)	-0.003 (-0.23)	-0.01 (-0.45)
DEALNUM3	0.12 (1.63) ^a	0.10 (0.99)	0.04 (2.11)	0.05 (1.93) ^a
DEALNUM4 +	0.19 (2.21) ^b	0.27 (1.90) ^a	0.03 (1.40)	0.06 (1.93) ^a
Table 8 controls	Yes	Yes	Yes	Yes
F-statistic	2.18 (0.02)	1.95 (0.03)	58.37 (0.00)	21.86 (0.00)

Table 11

Changes in performance of acquirer's own plants after acquisition.

The table reports the average total factor productivity (TFP) and operating margin in year -1 and the changes in TFP between year -1 and years $+1$, $+2$, and $+3$ for plants owned by firms that make an acquisition between 1980 and 2000 and have matching book-to-market data in Compustat. Acquirer data is identified from the SDC M&A database. Year 0 denotes the merger completion year. A plant is classified as Kept if it remains in the acquirer's possession as of year $+3$ and Sold if it is operating but not under the acquirer's ownership in year $+3$. We report statistics for two efficiency measures π : (1) Operating margin, which is ratio of the operating income before depreciation to the total plant shipments minus the industry median margin; (2) TFP, which is a plant's log output minus the predicted output based on a log-linear production function with squared and cross-product terms estimated for all plants in the industry. The operating margins and TFP statistics in this table are regression-adjusted for predictable time-series changes. Standard errors are in parentheses and the number of plants are below each performance statistic.

Full sample					Sample with acquirer BE/ME				
Panel A: Kept plants									
Statistic	π_{-1}	$\Delta\pi_{-1,+1}$	$\Delta\pi_{-1,+2}$	$\Delta\pi_{-1,+3}$	π_{-1}	$\Delta\pi_{-1,+1}$	$\Delta\pi_{-1,+2}$	$\Delta\pi_{-1,+3}$	
π = TFP	0.17 (21.13) ^a	0.07 (8.25) ^a	0.078 (15.60) ^a	0.069 (11.50) ^a	0.17 (21.13) ^a	0.07 (10.29) ^a	0.08 (9.75) ^a	0.069 (8.63) ^a	
N	15,290	15,290	15,290	15,290	9,362	9,362	9,362	9,362	
π = Margin	0.036 (22.44) ^a	0.005 (3.13) ^a	0.008 (6.15) ^a	0.01 (6.67) ^a	0.035 (17.50) ^a	0.007 (4.12) ^a	0.011 (5.79) ^a	0.017 (8.63) ^a	
N	15,426	15,426	15,426	15,426	9,398	9,398	9,398	9,398	
Panel B: Sold plants									
π = TFP	0.059 (4.47) ^a	0.006 (0.46)	0.02 (1.48)	0.036 (2.73) ^a	0.07 (3.89) ^a	-0.024 (-1.50)	-0.01 (-0.56)	0.035 (1.80)	
N	2,893	2,893	2,893	2,893	1,601	1,601	1,601	1,601	
π = Margin	0.007 (2.00) ^a	-0.008 (-2.85) ^a	-0.005 (-1.52)	-0.001 (-0.29)	0.01 (2.85) ^a	-0.02 (-4.50) ^a	-0.02 (-3.40) ^a	-0.008 (-1.70) ^a	
N	3,066	3,066	3,066	3,066	1,676	1,676	1,676	1,676	

t-statistics from test of significance of the average from zero in parentheses.

^a = significant at 1%; ^b = significant at 5%; ^c = significant at 10%.

We combine all the target and acquirer plants in one specification. We then estimate one specification with all plants and the lower panel reports data for sold plants. In each panel, we report results for the full sample and for

(a) Table 10: Repeat acquirers, disposition and performance changes.

(b) Table 11: Changes in performance of acquirer's own plants.

7 Short summary

- The paper studies post-merger restructuring using plant-level data for U.S. manufacturing acquisitions from 1981 to 2000.
- Acquirers sell or close nearly half of target plants within three years.
- The restructuring is much larger than normal industry-year plant turnover and much larger than restructuring after targeted partial-firm asset purchases.
- Related plants, target-main-division plants, and high-margin plants are more likely to be retained.
- Acquirers with higher skill in peripheral businesses retain more target plants.

- The interaction of acquirer skill and positive industry returns supports a comparative-advantage view of retention.
- Financial constraints matter: constrained acquirers are more likely to sell plants.
- Retained target plants improve in productivity and operating margins; sold target plants do not improve as much.
- Repeat acquirers and anti-takeover-law tests provide little evidence that real-asset outcomes are dominated by empire building.
- The main contribution is to show that mergers are followed by economically sensible boundary redrawing at the plant level.

8 Conclusion

8.1 Core conclusion

Maksimovic, Phillips, and Prabhala show that the boundaries of the firm are actively redrawn after mergers. In the three years after acquisition completion, acquirers sell or close almost half of the target plants they acquire. This post-merger restructuring is not random. It is linked to asset fit, target plant productivity, acquirer marginal skill, and industry conditions.

8.2 Economic intuition

The central intuition is that an acquisition transfers a bundle of assets, but the acquirer may have comparative advantage only over part of the bundle. After the transaction, the acquirer keeps assets for which its managerial and organizational capabilities are valuable. It sells assets that can be operated better elsewhere, especially when industry conditions make those assets valuable to other firms. It closes plants whose fundamentals are weak.

8.3 Incremental contribution revisited

The paper advances the M&A and theory-of-the-firm literatures in three major ways. First, it provides plant-level evidence on short-run post-merger restructuring, rather than relying only on long-run divestitures or firm-level performance. Second, it tests a comparative-advantage theory of firm boundaries by linking retention decisions to acquirer marginal skill and industry shocks. Third, it connects disposition decisions to real performance outcomes, showing that retained plants improve more than sold plants.

8.4 Implications for M&A and firm boundaries

The paper implies that merger evaluation should not stop at deal completion. The post-merger integration and divestiture process is central to value creation. Full-firm acquisitions may be rational even when acquirers want only part of the target, because the acquirer can subsequently sell or close mismatched assets. The firm boundary is therefore dynamic: mergers create an initial asset bundle, and restructuring refines it.

8.5 Limitations

The evidence is observational and cannot fully eliminate matching between acquirers and targets. The data are limited to manufacturing plants, so the conclusions may not fully generalize to intangible-asset or service industries. Plant-level measures exclude some headquarters-level costs. The paper also does not rule out overpayment or wealth transfers from acquirer shareholders to target shareholders. It shows that the operating side of post-merger asset allocation looks economically sensible, not that every merger creates shareholder value for acquirers.

8.6 Takeaway

The enduring takeaway is that post-merger restructuring is a major mechanism through which firms redefine their boundaries. The evidence is hard to reconcile with pure empire building. Acquirers generally retain plants where they have comparative advantage and improve those plants after acquisition. Mergers are therefore best understood not as a final ownership allocation, but as the beginning of a rapid real-asset reallocation process.